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Preface / 前言

This minimal working example demonstrates the typesetting architecture of the Bilingual Academic Textbook Skill.

本最小工作范例用于展示双语学术教材排版 Skill 的核心架构与交互特性。

Chapter 1

Measure Theory / 测度论

> *"The sets whose measure we can define* ***by virtue of*** *the preceding ideas we will call* ***measurable sets****."* — È. Borel, 1898
>
> "依据前述思想我们能够定义其测度的集合，我们将称之为**可测集**。"
> ——E. 博雷尔，1898 年

[Orig. p. 1]

by virtue of
依据 / 凭借
高频数学论证学术表达

measurable set
可测集
依据外测度能赋予测度的集合

This chapter is devoted to the construction of Lebesgue measure in $\mathbb{R}^d$ and the study of measurable functions. We begin with the definition of **exterior measure** for any subset $E \subset \mathbb{R}^d$. This concept is formulated by approximating E with coverings of open boxes or cubes.

本章致力于在 $\mathbb{R}^d$ 中建立勒贝格测度，并深入研究可测函数。我们首先引入 $\mathbb{R}^d$ 中任意子集 E 的外测度定义。该概念是通过用开方块或立方体覆盖来逼近 E 而建立的。

exterior measure
外测度
全空间子集上的广义外测度

The concept of measurable functions is a natural outgrowth of the idea of measurable sets. Furthermore, the class of measurable functions has the notable property of being closed under taking **pointwise limits**.

可测函数的概念是可测集思想的自然延伸。此外，可测函数类具有一个显著特性：在取逐点极限运算下是封闭的。

pointwise limit
逐点极限
函数序列在每个点处的极限

1.1 Exterior Measure / 外测度

[Orig. p. 2]

exterior measure
外测度
集合 E 的外测度

Definition 1.1. If E is any subset of $\mathbb{R}^d$, the **exterior measure** of E, denoted by $m_*(E)$, is defined as

$$m_*(E) = \inf \sum_{j=1}^{\infty} |Q_j|,$$

where the infimum is taken over all countable coverings of E by closed cubes $\{Q_j\}_{j=1}^{\infty}$.

若 E 为 $\mathbb{R}^d$ 的任意子集，则 E 的**外测度**（记为 $m_*(E)$）定义为

$$m_*(E) = \inf \sum_{j=1}^{\infty} |Q_j|,$$

其中下确界取遍所有用闭立方体序列 $\{Q_j\}_{j=1}^{\infty}$ 对 E 进行的可数覆盖。

Theorem 1.2 (Countable Sub-additivity / 可数次可加性). *If* $E = \bigcup_{j=1}^{\infty} E_j$, *then*

$$m_*(E) \leq \sum_{j=1}^{\infty} m_*(E_j).$$

若 $E = \bigcup_{j=1}^{\infty} E_j$，则

$$m_*(E) \leq \sum_{j=1}^{\infty} m_*(E_j).$$

Proof. If $m_*(E_j) = \infty$ for some j, the inequality is trivially true. Otherwise, for every $\epsilon > 0$ and each j, choose a sequence of cubes $\{Q_{j,k}\}_k$ covering E_j such that

$$\sum_{k=1}^{\infty} |Q_{j,k}| \leq m_*(E_j) + \frac{\epsilon}{2^j}.$$

Then the collection of all cubes $\{Q_{j,k}\}_{j,k}$ covers E, so that

$$m_*(E) \leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} |Q_{j,k}| \leq \sum_{j=1}^{\infty} m_*(E_j) + \epsilon.$$

Since $\epsilon > 0$ is arbitrary, the result follows.

若对某个 j 有 $m_*(E_j) = \infty$，则不等式平凡成立。否则，对任意 $\epsilon > 0$ 以及每个 j，选取覆盖 E_j 的立方体序列 $\{Q_{j,k}\}_k$，使得

$$\sum_{k=1}^{\infty} |Q_{j,k}| \leq m_*(E_j) + \frac{\epsilon}{2^j}.$$

于是全体立方体族 $\{Q_{j,k}\}_{j,k}$ 构成 E 的覆盖，由此可知

$$m_*(E) \leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} |Q_{j,k}| \leq \sum_{j=1}^{\infty} m_*(E_j) + \epsilon.$$

由于 $\epsilon > 0$ 的任意性，结论即告成立。

□

Academic Glossary / 学术术语总表

English Term / 学术用语	Chinese Definition / 中文规范释义	Academic Context / 语境注记
almost disjoint	几乎不相交	交集测度为零的几何族
by virtue of	依据 / 凭借	高频数学论证学术表达
exterior measure	外测度	全空间子集上的广义外测度
measurable set	可测集	依据外测度能赋予测度的集合
pointwise limit	逐点极限	函数序列在每个点处的极限